Brandon K Markham
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Education

Texas State University
Bachelor of Science in Electrical and Computer Engineering
January 2022 – May 2025
GPA: 3.4

Skills

System Validation and Testing:
Bench-Level Validation Testing, Automated Test Procedure Execution, Failure Analysis, Analog and Digital Circuit Debugging, System-Level Diagnostics, Board-Level Troubleshooting, Audio and Power Electronics Fundamentals

Software & Tools:
Python, C/C++, LTspice, KiCad, Fusion 360, MATLAB(Academic), Microsoft Excel

Lab & Test Equipment:
Oscilloscopes, Power Supplies, Signal Generators, Digital Multimeters, Automated Test Fixtures

Engineering Experience

Engineering Technician – Applied Materials February 2026 - Present

- Test and troubleshoot system-level and component-level issues in mechanical, pneumatic, and vacuum subsystems on semiconductor manufacturing equipment
- Perform new system builds in support of production operations.
- Interpret schematics, mechanical diagrams, and engineering documentation to perform diagnostics and ensure system functionality.
- Execute functional, process, and safety tests on Applied Materials platforms in accordance with workmanship standards and engineering specifications.
- Generate Quality Notifications and detailed test documentation to support root cause analysis and continuous improvement initiatives.
- Maintain accurate logbooks and procedural records to ensure traceability, compliance, and operational consistency.
- Follow strict safety and quality standards while working in controlled manufacturing environments.

Founder / Analog Design Engineer – Bat City Sound Company January 2024 - Present

- Designed and simulated analog audio circuits including gain stages, clipping networks, filters, and impedance-matching stages
- Performed circuit modeling and frequency-domain analysis using LTspice to validate gain structure, stability, and noise performance.
- Breadboarded, tested, and iterated hardware prototypes using lab instrumentation
- Developed PCB layouts in KiCad and coordinated enclosure, power, and grounding strategies for low-noise audio performance

Embedded Systems Engineer – Bat City Sound Company January 2024 - Present

- Built a digital guitar looper pedal on the STM32 platform with real-time audio playback
- Verified low-latency audio performance while balancing SD card storage efficiency
- Implemented undo/redo limited to overdub layers while preserving the base recording
- Designed save/load functionality in WAV format with reliable SDMMC + FatFS performance
- Developed hardware I/O JFET buffers for impedance matching

Software Engineer – Bat City Sound Company January 2024 – Present

- Developed a Python-based inventory management application to manage parts, quantities, usage history, and reorder thresholds
- Automated inventory updates based on production and build activity to reduce manual tracking errors
- Designed data structures and workflows to support component-level traceability across multiple product builds
- Implemented reporting and data export features to support purchasing, planning, and cost analysis
- Maintained and iterated the system based on evolving operational requirements